

Clinical Risk Assessment Report: Patient 85

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 33.3% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.50x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +6.9%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.35 [Bottom 12%] (+6.9%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 1.11 [Top 9%] (+4.0%)**
Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 2.18 [Top 2%] (+2.9%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.74 [Avg (72th %ile)] (+2.3%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT) **Value: 1.06 [Top 13%] (+1.8%)**
Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.

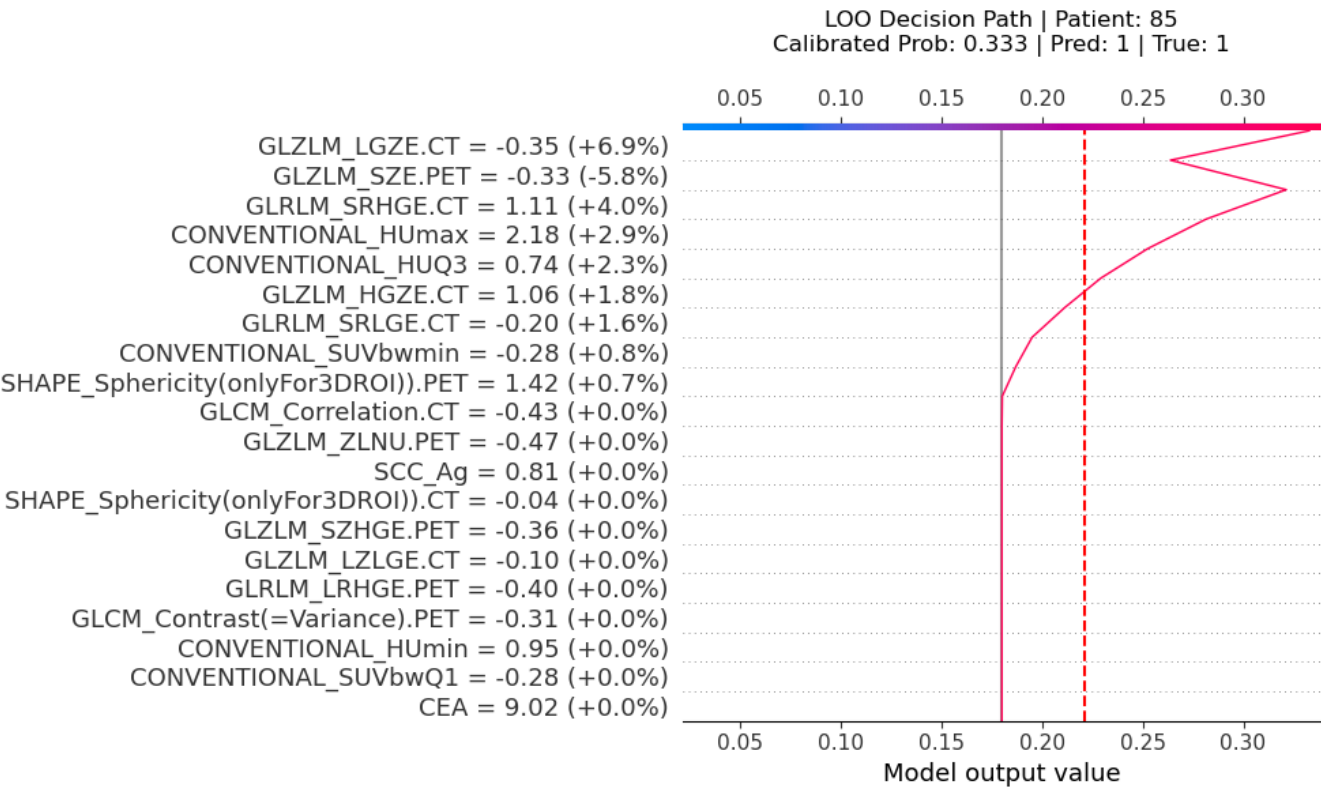
Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.2 [Bottom 21%] (+1.6%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: -0.28 [Avg (47th %ile)] (+0.8%)**
Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.33 [Avg (31th %ile)] (-5.8%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

